

Programme	B. Sc. Computer Science				
Course Title	Advanced Python for Data Science				
Type of Course	Elective				
Semester	VI				
Academic Level	300 ■ 399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	1. Data Science Concepts 2. Python basics				
Course Summary	This course provides insight into the basic concepts of Python required for Data Science. It includes array fundamentals, array transformations, and matrices fundamentals. The analysis of data using Pandas will help the students to understand the basics of data analysis				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the concepts of arrays, matrices and their transformations	U	C	Instructor-created exams / Quiz
CO2	Create informative plots using Python packages	Ap	P	Modelling Assignments/ Case Studies
CO3	Understand the loading mechanism of different types of data and manipulate them	U	C	Instructor-created exams / Quiz
CO4	Analyse the data using Pandas and Data Frames	An	P	Modelling Assignments/ Case Studies
CO5	Understand the concepts of random tensors and generate tensors from various distributions	U	C	Instructor-created exams / Quiz
CO6	Familiarize with various TensorFlow operations needed for Data Science	U	C	Instructor-created exams / Quiz

* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C)
 # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
I	Arrays, Matrix manipulation using NumPy		10	12
	1	Array creation, sorting, concatenating	2	
	2	Shape and size of an array, basic arithmetic operations on an array, broadcasting	2	
	3	Aggregate functions on arrays, Unique and count operations	2	
	4	Matrices using NumPy	2	
	5	Transpose, reverse, flatten and ravel	2	
II	Data Analysis and Manipulation using Pandas		12	18
	6	Series - constructing from an array, using explicitly defined indices, using a dictionary.	2	
	7	Data Frame - constructing from arrays, dictionaries, structured arrays, and series, Indexing of data frames	3	
	8	Arithmetic and Binary operations on Data frame	3	
	9	Broadcasting operations	2	
	10	Universal functions, melt() and pivot()	2	
III	Other Python packages for data science		10	14
	11	Scipy, Scikit-learn, PyTorch, Seaborn, Scrapy, and Beautiful Soup.	3	
	12	Python Data Operations: Importing and Exporting Data, Data Cleansing	3	
	13	Processing CSV Data, Processing JSON Data, Processing XLS Data.	2	
	14	Data Analysis: Measuring Central Tendency, Measuring Variance, and Correlation in Python	2	
IV	TensorFlow Fundamentals		16	26
	15	Tensors, creation of tensors and random tensors, Tensors from the Normal distribution, Poisson distribution, set_seed()	2	
	16	Tensor attributes, size, rank and reshaping of a tensor	2	
	17	Tensor arithmetic, relational, logical operations. Shuffle()	2	
	18	Reduce operations on tensor Dimension-wise	2	
	19	Ragged tensors, TensorArray, dynamic arrays,	2	

	20	unique(), fill(), concat(), gather(), ones(), ones_like(), zeros(),	2	
	21	eye(), range(), repeat, reverse(), roll(), slice(), sort(),	2	
	22	split(),squeeze(), tile(), stack(), unstack(), tensordot()	2	
V	Open ended module - Hands-on Data Structures:		12	
	Practical Applications, Case Study and Course Project			
	1	Use Pandas and NumPy to efficiently process and analyze CSV, Excel, or JSON data	4	
	2	Create compelling visual insights using Matplotlib, Seaborn, or Plotly	3	
	3	Case studies with Tensor flow	5	

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	1	-	-	2	2	2						
CO 2	-	1	-	-	2	2						
CO 3	-	-	2	-	2	2						
CO 4	-	1	1	2	2	2						
CO 5	1	-	-	-	2	2						
CO 6	-	-	2	2	2	2						

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓			✓
CO 2	✓	✓	✓	✓
CO 3	✓		✓	✓
CO 4	✓	✓	✓	✓
CO 5	✓			✓
CO 6	✓			✓

References

1. VanderPlas, Jake. Python Data Science Handbook: Essential Tools for Working with Data. United States, O'Reilly Media, 2016.
2. Rogel-Salazar, Jesus. Data Science and Analytics with Python. United Kingdom, CRC Press, 2018.
3. <https://numpy.org/doc/>
4. <https://pandas.pydata.org/docs/>
5. <https://www.tensorflow.org/guide>